

A Two-Scale Prime-Square Tail Expansion with a Uniform Cubic Remainder

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Abstract

We stay inside the fixed finite-clock, universally distance-two-safe phasewise lag-two class in which the interpolation coefficient $c_{11}(r)$ vanishes at every phase. RH-381 expresses the square-clock gap as an exact infinite increment sum. Let

$$a_{j+1} = \frac{1}{p_{j+1}^2 - 1}, \quad T_y = \sum_{j \geq y} a_{j+1}, \quad S_y = \sum_{j \geq y} a_{j+1}^2,$$

where $3 = p_1 < p_2 < \dots$ are the odd primes. For $e_m = \prod_{p \text{ odd}} (1 - m/p^2)$ and $u_m = e_m/e_1$, put

$$\begin{aligned} X_\infty &= 2u_4 - 4u_5 + 6u_6 - 8u_7 + 10u_8, \\ Y_\infty &= 6u_4 - 16u_5 + 30u_6 - 48u_7 + 70u_8, \\ m_\infty &= 2u_3 - 4u_4 + 6u_5 - 8u_6 + 10u_7 - 12u_8. \end{aligned}$$

All-order product inequalities, rather than finite fitting, give for every $y \geq 1$

$$B_\infty - G(q_y) = \frac{2X_\infty}{\pi^2} T_y + \frac{Y_\infty + 2m_\infty}{\pi^2} T_y^2 + \frac{Y_\infty - 2m_\infty}{\pi^2} S_y + R_y,$$

with the explicit stronger ledger

$$|R_y| \leq \frac{3301}{6\pi^2} T_y^3 < \frac{551}{\pi^2} T_y^3.$$

The numerator contributes 931/2 and the memory channel 254/3. The proof keeps S_y independent, treats $R_8 = P_y E_8$ separately, uses $E_9 = 0$ exactly, and never introduces E_{10} . No prime number theorem, p_y -scale rewrite, growing clock, adaptive-capacity limit, operator, trace formula, zero identification, or implication for the Riemann hypothesis is supplied.

Keywords: Möbius function; prime-square Euler tail; two-scale asymptotics; finite clocks; explicit remainder.

1 Frozen class and exact gap input

The quantifiers are inherited without enlargement from RH-379–RH-381 [1, 3, 2].

Definition 1.1 (Fixed-clock phasewise class). Fix an integer $q \geq 1$ before taking $N \rightarrow \infty$. At every $r \in \mathbb{Z}/q\mathbb{Z}$, choose

$$f_r : \{-1, 0, 1\}^2 \longrightarrow \{-1, +1\}, \quad \epsilon_n = f_{n \bmod q}(\mu(n-2), \mu(n)).$$

The family is universally distance-two-safe if no ternary input can give $\epsilon_n = \epsilon_{n+2} = +1$. It belongs to the present class only when the RH-379 interpolation coefficient $c_{11}(r)$ of $\mu(n-2)\mu(n)$ is zero at every phase. Write $G(q)$ for the exact fixed-clock optimum in this class.

No phase cancellation is assumed. If $c_{11}(r)$ is active, a phase-weighted shift-two Möbius correlation appears and is not controlled by the frozen inputs.

Let $3 = p_1 < p_2 < \dots$ be the odd primes and define

$$P_j = \prod_{i \leq j} p_i^2, \quad q_j = 4P_j, \quad A_j = \prod_{i \leq j} (p_i^2 - 1), \quad D_j = \prod_{i \leq j} (p_i^2 - 2). \quad (1)$$

If $R_\ell^{(j)}$ is the RH-374 positive-run count, put

$$\mathcal{E}_j = R_2^{(j)} + R_4^{(j)} + R_6^{(j)} + R_8^{(j)}, \quad L_j = 2R_2^{(j)} + 4R_4^{(j)} + 6R_6^{(j)} + 8R_8^{(j)}, \quad (2)$$

$$M_j = 2R_3^{(j)} + 4R_5^{(j)} + 6R_7^{(j)}, \quad X_j = \frac{L_j - 2\mathcal{E}_j}{A_j}. \quad (3)$$

RH-380 proves the exact increment, and RH-381 telescopes it after all fixed-clock limits have been taken [3, 2]:

Proposition 1.2 (Frozen exact infinite increment sum). *For every $y \geq 1$,*

$$B_\infty - G(q_y) = \sum_{j \geq y} \left[\frac{2X_j}{\pi^2} a_{j+1} + \frac{M_j}{A_j} a_{j+1} \left(\frac{4}{\pi^2} - H_{j+1} \right) \right], \quad (4)$$

where

$$a_{j+1} = \frac{1}{p_{j+1}^2 - 1}, \quad H_j = \kappa_2 \frac{A_j}{D_j}.$$

Equation (4) is an input, not a simultaneous limit in N and j . Its use below creates no clock $q = q(N)$.

2 Finite Euler ratios and the terminal run

For $1 \leq m \leq 9$, define

$$E_m^{(j)} = \prod_{i \leq j} \left(1 - \frac{m}{p_i^2} \right), \quad U_m^{(j)} = \frac{E_m^{(j)}}{E_1^{(j)}} \quad (1 \leq m \leq 8), \quad (5)$$

and let

$$e_m = \prod_{p \text{ odd}} \left(1 - \frac{m}{p^2} \right), \quad u_m = \frac{e_m}{e_1} \quad (1 \leq m \leq 8). \quad (6)$$

These products converge and are positive for $m \leq 8$ because $\sum_{n \geq 3} n^{-2} < \infty$ and every factor is positive.

The terminal convention matters. RH-374 proves [4]

$$R_\ell^{(j)} = P_j (E_\ell^{(j)} - 2E_{\ell+1}^{(j)} + E_{\ell+2}^{(j)}), \quad 1 \leq \ell \leq 7, \quad (7)$$

but the last run is

$$R_8^{(j)} = P_j E_8^{(j)}. \quad (8)$$

Thus (7) is not extended to $\ell = 8$. In particular, E_{10} is neither needed nor defined in the run ledger.

Lemma 2.1 (Exact numerator and memory Euler forms). *For every $j \geq 1$,*

$$X_j = 2U_4^{(j)} - 4U_5^{(j)} + 6U_6^{(j)} - 8U_7^{(j)} + 10U_8^{(j)}, \quad (9)$$

$$\frac{M_j}{A_j} = 2U_3^{(j)} - 4U_4^{(j)} + 6U_5^{(j)} - 8U_6^{(j)} + 10U_7^{(j)} - 12U_8^{(j)}. \quad (10)$$

Moreover $0 \leq M_j/A_j \leq 1$.

Proof. The combinatorial identity

$$L_j - 2\mathcal{E}_j = 2R_4^{(j)} + 4R_6^{(j)} + 6R_8^{(j)}$$

uses the terminal term (8) separately. Substitution of (7) only at lengths four and six gives (9); no E_9 or E_{10} appears.

For the memory statistic, substitute (7) at lengths three, five, and seven. Before simplification, the numerator is

$$2E_3^{(j)} - 4E_4^{(j)} + 6E_5^{(j)} - 8E_6^{(j)} + 10E_7^{(j)} - 12E_8^{(j)} + 6E_9^{(j)}.$$

Here

$$E_9^{(j)} = 0$$

exactly for every $j \geq 1$, because its $p_1 = 3$ factor is $1 - 9/9$. Division by $E_1^{(j)}$ proves (10). Finally, M_j counts at most all positive sites in the support word, whose number is A_j . $\square$

Passing to the infinite products in these two formulas defines the three coefficients used in the theorem:

$$X_\infty = 2u_4 - 4u_5 + 6u_6 - 8u_7 + 10u_8, \quad (11)$$

$$Y_\infty = 6u_4 - 16u_5 + 30u_6 - 48u_7 + 70u_8, \quad (12)$$

$$m_\infty = 2u_3 - 4u_4 + 6u_5 - 8u_6 + 10u_7 - 12u_8. \quad (13)$$

Since $0 \leq M_j/A_j \leq 1$ and the Euler ratios converge,

$$0 \leq m_\infty \leq 1. \quad (14)$$

3 An all-order product expansion

Put

$$T_j = \sum_{k \geq j} a_{k+1}, \quad S_j = \sum_{k \geq j} a_{k+1}^2. \quad (15)$$

The elementary comparison

$$T_y \leq \sum_{n > p_y} \frac{1}{n^2 - 1} = \frac{1}{2} \left(\frac{1}{p_y} + \frac{1}{p_y + 1} \right)$$

shows convergence and $T_y \rightarrow 0$. This is not a prime-counting asymptotic.

Lemma 3.1 (Bonferroni product bounds). *Let $0 \leq b_i < 1$ and $x = \sum_i b_i < \infty$. Then*

$$x - \frac{x^2}{2} \leq 1 - \prod_i (1 - b_i) \leq x, \quad \prod_i (1 - b_i) \leq \frac{1}{1 + x}. \quad (16)$$

Proof. For a finite product, the union bound gives the upper inequality. Keeping the first two elementary symmetric sums gives

$$1 - \prod_i (1 - b_i) \geq \sum_i b_i - \sum_{i < k} b_i b_k \geq x - \frac{x^2}{2}.$$

Also $(1 - b_i)^{-1} \geq 1 + b_i$, so

$$\prod_i (1 - b_i)^{-1} \geq \prod_i (1 + b_i) \geq 1 + x.$$

Finite truncation followed by monotone passage proves all three inequalities for the infinite product. No Taylor truncation is being treated as an identity. $\square$

Lemma 3.2 (Uniform Euler-ratio expansion). *For $3 \leq m \leq 8$ and every $j \geq 1$,*

$$0 \leq U_m^{(j)} - u_m - (m-1)u_m T_j \leq U_m^{(j)}(m-1)^2 T_j^2 \leq \frac{9-m}{8}(m-1)^2 T_j^2. \quad (17)$$

Proof. Since $j \geq 1$, every tail prime is at least $p_{j+1} \geq 5$. Hence, for $3 \leq m \leq 8$,

$$0 \leq (m-1)a_{k+1} \leq \frac{7}{24} < 1 \quad (k \geq j),$$

so Lemma 3.1 applies to every factor below. Factorwise cancellation gives

$$P_{m,j} := \frac{u_m}{U_m^{(j)}} = \prod_{k \geq j} (1 - (m-1)a_{k+1}). \quad (18)$$

Write $x = (m-1)T_j$ and $\delta = 1 - P_{m,j}$. Lemma 3.1 gives

$$P_{m,j} \leq \frac{1}{1+x}, \quad \frac{x}{1+x} \leq \delta \leq x.$$

Because $u_m = U_m^{(j)} P_{m,j}$,

$$U_m^{(j)} - u_m - x u_m = U_m^{(j)} [(1+x)\delta - x].$$

The last bracket lies between zero and x^2 . Finally, $U_m^{(j)}$ contains the $p=3$ ratio $(9-m)/8$, while every later factor lies in $(0,1)$, proving the last bound. $\square$

Proposition 3.3 (Numerator and memory first variations). *For every $j \geq 1$,*

$$|X_j - X_\infty - Y_\infty T_j| \leq \frac{931}{4} T_j^2, \quad (19)$$

$$\left| \frac{M_j}{A_j} - m_\infty \right| \leq 63 T_j. \quad (20)$$

Proof. Apply Lemma 3.2 to (9). The linear coefficient is exactly (12), and the absolute quadratic ledger is

$$\frac{45}{4} + 32 + \frac{225}{4} + 72 + \frac{245}{4} = \frac{931}{4}.$$

For (20), the union-bound consequence

$$0 \leq U_m^{(j)} - u_m \leq U_m^{(j)}(m-1)T_j \leq \frac{9-m}{8}(m-1)T_j$$

is applied to (10). The six absolute contributions are

$$3 + \frac{15}{2} + 12 + 15 + 15 + \frac{21}{2} = 63.$$

Both estimates hold for the entire infinite tail, not just the finite rows in the certificate. $\square$

4 The H loss and exact tail algebra

RH-381 derives the exact product [2]

$$\frac{H_{j+1}}{4/\pi^2} = \prod_{k \geq j+1} (1 - a_{k+1}). \quad (21)$$

Define its normalized loss

$$d_{j+1} := 1 - \frac{H_{j+1}}{4/\pi^2}. \quad (22)$$

Lemma 3.1, now with $x = T_{j+1}$, gives

$$d_{j+1} = T_{j+1} + \eta_{j+1}, \quad -\frac{1}{2}T_{j+1}^2 \leq \eta_{j+1} \leq 0. \quad (23)$$

Equivalently,

$$\left| \left(\frac{4}{\pi^2} - H_{j+1} \right) - \frac{4}{\pi^2} T_{j+1} \right| \leq \frac{2}{\pi^2} T_{j+1}^2.$$

We next retain the two different quadratic scales.

Lemma 4.1 (Quadratic and cubic telescopes). *For every $y \geq 1$,*

$$\sum_{j \geq y} a_{j+1} T_j = \frac{T_y^2 + S_y}{2}, \quad (24)$$

$$\sum_{j \geq y} a_{j+1} T_{j+1} = \frac{T_y^2 - S_y}{2}. \quad (25)$$

If $C_y = \sum_{j \geq y} a_{j+1}^3$, then

$$\sum_{j \geq y} a_{j+1} T_j T_{j+1} = \frac{T_y^3 - C_y}{3} \leq \frac{T_y^3}{3}, \quad (26)$$

$$\sum_{j \geq y} a_{j+1} T_{j+1}^2 \leq \frac{T_y^3}{3}, \quad (27)$$

$$\sum_{j \geq y} a_{j+1} T_j^2 \leq T_y^3. \quad (28)$$

Proof. Since $T_j = T_{j+1} + a_{j+1}$,

$$\begin{aligned} a_{j+1} T_j &= \frac{1}{2}(T_j^2 - T_{j+1}^2 + a_{j+1}^2), \\ a_{j+1} T_{j+1} &= \frac{1}{2}(T_j^2 - T_{j+1}^2 - a_{j+1}^2). \end{aligned}$$

Finite summation and $T_j \rightarrow 0$ prove (24)–(25). For the mixed cube,

$$T_j^3 - T_{j+1}^3 = 3a_{j+1} T_j T_{j+1} + a_{j+1}^3,$$

which proves (26). The alternative expansion

$$T_j^3 - T_{j+1}^3 = 3a_{j+1} T_{j+1}^2 + 3a_{j+1}^2 T_{j+1} + a_{j+1}^3$$

proves (27); nonnegative finite terms justify passage to the limit. Finally $T_j \leq T_y$ and $\sum_{j \geq y} a_{j+1} = T_y$ give (28). $\square$

5 The two-scale theorem

Theorem 5.1 (Two-scale prime-square tail expansion). *For every $y \geq 1$ in the fixed class of Definition 1.1,*

$$\boxed{B_\infty - G(q_y) = \frac{2X_\infty}{\pi^2}T_y + \frac{Y_\infty + 2m_\infty}{\pi^2}T_y^2 + \frac{Y_\infty - 2m_\infty}{\pi^2}S_y + R_y,} \quad (29)$$

where

$$\boxed{|R_y| \leq \frac{3301}{6\pi^2}T_y^3 < \frac{551}{\pi^2}T_y^3.} \quad (30)$$

Proof. Write

$$X_j = X_\infty + Y_\infty T_j + \rho_j, \quad |\rho_j| \leq \frac{931}{4}T_j^2$$

by Proposition 3.3. The numerator part of (4) is therefore

$$\sum_{j \geq y} \frac{2a_{j+1}X_j}{\pi^2} = \frac{2X_\infty}{\pi^2}T_y + \frac{Y_\infty}{\pi^2}(T_y^2 + S_y) + R_y^{(X)}, \quad (31)$$

where Lemma 4.1 gives

$$|R_y^{(X)}| \leq \frac{931}{2\pi^2} \sum_{j \geq y} a_{j+1}T_j^2 \leq \frac{931}{2\pi^2}T_y^3. \quad (32)$$

Let $m_j = M_j/A_j$. By (22), the memory part is

$$\frac{4}{\pi^2} \sum_{j \geq y} a_{j+1}m_j d_{j+1}.$$

Subtract its proposed main term $(4m_\infty/\pi^2) \sum a_{j+1}T_{j+1}$. Equations (20), (14), and (23) give

$$\begin{aligned} |m_j d_{j+1} - m_\infty T_{j+1}| &\leq |m_j - m_\infty|T_{j+1} + m_j|d_{j+1} - T_{j+1}| \\ &\leq 63T_j T_{j+1} + \frac{1}{2}T_{j+1}^2. \end{aligned}$$

Using (25), (26), and (27), the memory channel becomes

$$\sum_{j \geq y} a_{j+1}m_j \left(\frac{4}{\pi^2} - H_{j+1} \right) = \frac{2m_\infty}{\pi^2}(T_y^2 - S_y) + R_y^{(M)}, \quad (33)$$

where

$$|R_y^{(M)}| \leq \frac{4}{\pi^2} \left(63 \frac{T_y^3}{3} + \frac{1}{2} \frac{T_y^3}{3} \right) = \frac{254}{3\pi^2}T_y^3. \quad (34)$$

Adding (31) and (33) produces the opposite S_y signs in (29). The explicit budget is

$$\frac{931}{2} + \frac{254}{3} = \frac{3301}{6} = 550 + \frac{1}{6} < 551,$$

which proves (30). $\square$

Remark 5.2 (Why S_y cannot be suppressed). The numerator uses $\sum a_{j+1}T_j = (T_y^2 + S_y)/2$, whereas the memory channel uses $\sum a_{j+1}T_{j+1} = (T_y^2 - S_y)/2$. No relation replacing S_y by a fixed multiple of T_y^2 is assumed or proved. Dropping this scale would erase an exact sign distinction.

6 Exact artifact and a scoped sign mutation

The standard-library artifact implements only exact rational arithmetic for the theorem ledger. It checks 24 finite Euler-ratio product rows, four Bonferroni rows, four quadratic/cubic telescope rows, and four finite endpoint gap rows. These rows reproduce the symbolic identities and exercise their bounds; the proof of Theorem 5.1 is the all-order argument above.

The certificate also freezes one deliberately scoped mutation. Truncate the Euler endpoint after the nineteenth odd prime, $p_{19} = 71$, and compare the one-prime tail from $p_{18} = 67$. Thus

$$a = T = \frac{1}{71^2 - 1} = \frac{1}{5040}, \quad S = a^2 = \frac{1}{5040^2}.$$

All endpoint coefficients are exact fractions. Under the displayed theorem sign, the exact residual divided by $551a^3$ is

$$0.042746686479386 \dots < 1.$$

Mutate only the memory term

$$-2m_{19}S \quad \mapsto \quad +2m_{19}S,$$

while leaving the numerator term $+Y_{19}S$ unchanged. The approximation then changes by exactly $4m_{19}S$, and the exact rational residual ratio is

$$7.335622869337969 \dots > 1.$$

This is a reproduction-only wrong-sign rejection. It is not a regression, not evidence for the infinite theorem, and not the different mutation that would flip the numerator's YS term.

The result builder verifies 33 immutable predecessor files against both their live SHA-256 digests and the exact blobs at the declared releases: seven from RH-374, eight each from RH-379, RH-380, and RH-381, and two from the RH-MVP2 archive. Mutable root policy and handoff files are excluded. The result has a recursively closed Draft 2020-12 schema. Result and archive loaders reject duplicate keys, non-finite constants, unsafe paths, membership drift, release rebinding, and hash drift.

Reproduction from this directory is

```
make result
make schema
make test
make pdf
make archive
```

7 Scope and limitations

Theorem 5.1 is a complete second-order theorem inside one frozen class. It is not a statement about a larger dynamical or arithmetic system.

- Every clock is finite and fixed before $N \rightarrow \infty$. There is no diagonal choice $q = q(N)$.
- Universal safety and phasewise $c_{11}(r) = 0$ are hypotheses. Active c_{11} remains blocked by a phase-weighted shift-two correlation theorem not present in the source lock.
- The independent S_y scale and its opposite memory sign are part of the theorem. Neither is replaced by numerical fitting.
- No prime number theorem is used, and the result is not rewritten as an asymptotic in p_y .

- No adaptive-capacity convergence, geometrically selected non-Parry measure, or deterministic strong-space Ulam bridge is inferred.
- The work constructs no canonical intrinsic dynamical spectral determinant, scattering completion, self-adjoint generator, von Mangoldt prime-power trace identity, or completed-zeta divisor equality. Gates A–E remain false/open [5].
- There is no Hilbert–Pólya operator, no identification of Riemann zeros, and no implication for the Riemann hypothesis.

Declarations

Data and code availability. All theorem inputs, exact source code, tests, closed result schema, source locks, and archive manifests are contained in the repository paper directory. No private dataset is used.

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AI assistance disclosure. AI-assisted tools supported source triage, symbolic checking, deterministic artifact generation, drafting, and adversarial review. Every released claim is constrained by immutable repository sources and reproducible verification artifacts; responsibility for the scoped mathematical statements remains with the research program.

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